

VIGNESHWAR MURUGASAMY

Kokomo, IN | (269) 341 1315 | dr.vigneshwar@gmail.com

Manufacturing Process Engineer | Semiconductor & Electronics Manufacturing | Equipment, Production

Manufacturing Engineer with 7+ years of experience in powertrain, gear machining, and high-volume manufacturing environments, specializing in process development, production ramp-up, and NPI support. Strong background in Lean Manufacturing and Six Sigma methodologies, including 8D, 5-Why, DMAIC, and CAPA, with hands-on experience driving process stability, variability reduction, and continuous improvement. Skilled in supporting PPAP, FMEA, and DOE activities while collaborating with cross-functional teams to improve safety, quality, and throughput in automotive manufacturing systems.

SKILLS

Software: AutoCAD, SolidWorks, Power BI, STAAD.Pro, Minitab, MPRQ, SAP, MS Excel & PowerPoint, Pro-E

Equipment & Operations Support: Tool de-installation support, equipment installation support, factory floor modifications, system commissioning, equipment troubleshooting, production changeovers, validation testing, and safe equipment handling in high-volume manufacturing environments

Safety & Compliance: Lockout/Tagout (LOTO) execution and verification, EHS regulatory compliance (OSHA-based practices), machine energy isolation procedures, safe equipment shutdown/startup protocols, tool de-installation safety planning, risk assessment (JSA/JHA), ergonomic risk evaluation and workstation safety improvements, lifting/rigging safety awareness, confined space and hazardous energy control (where applicable), factory floor hazard identification and mitigation, and compliance with manufacturing safety audits and standard work procedures.

Lean Methodologies: JIT, Six Sigma (5s), FMEA, RFPs, SPC, Hoshin Kanri, PDCA, APQP, Lean, Value Stream Mapping, PPAP, Kaizen, ISO/ATF Standards, TPM, QMS, GD&T

Engineering & Process Skills: Manufacturing process validation, equipment testing, yield improvement, production scaling, system integration, EOL testing, defect analysis, manufacturing capability assessment, DFMA (Design for Manufacturing & Assembly), automation support

EDUCATION

Master of Science, Industrial & Systems Engineering, Northern Illinois University, DeKalb, IL

December 2023

Bachelor of Engineering, Mechanical Engineering, Anna University, India

May 2016

PROFESSIONAL EXPERIENCE

Process Manufacturing Engineer Stellantis, (contract)Kokomo,

June 2024 – May 2026

- Conducted process assessments for assembling EDM E-Motor offset and powertrain (Board, Stator & Rotor) components (800V, 400V), identifying key improvement areas.
- Owned end-to-end delivery of 4+ cross-functional global operational projects, defining goals, success metrics, and execution plans, delivering 3 programs on schedule within planned scope.
- Leading DVP&R tooling de-installation entire I4 EVO electrical engine line to accommodate the H4 and T4 engines at the line including shutdown planning, utility disconnects, and physical removal of automated systems in a high-volume manufacturing environment.
- Applied LOTO (Lockout/Tagout) procedures during robot stations and safe guarded stations de-energizing and servicing, ensuring safe execution of tool removal machining activities with zero safety violations.
- Managed factory floor equipment moves and line modifications with cross-functional teams, including controls engineers and contractors, to successfully integrate a new model (NM) into the model manager/scheduler system.
- Led continuous improvement initiatives for EDM, stator, rotor, PIM board assembly and SoC manufacturing lines, driving process optimization, cycle time reduction, throughput improvements, and production readiness activities.
- Supported development and deployment of manufacturing operating systems through tiered daily management, KPI reviews, escalation tracking, supporting corrective action and standardized work practices across production teams.
- Established KPI tracking and performance dashboards for SQDC metrics, throughput, uptime, cycle time, quality performance, and launch readiness supporting operational decision making.
- Led electronics-style fully automated manufacturing environments, supporting assembly, test stations, conveyors, and semi-automated production equipment in high-throughput operations.
- Performed manufacturing system analysis including capacity planning, line balancing, workflow optimization, process improvements, and production layout evaluations to improve efficiency and space utilization.

- Led Kaizen and continuous improvement activities focused on reducing waste, improving process flow, and increasing Smart cell machines in smart factory combining AI tools & IO.
- Supported implementation of digital performance monitoring through Power BI dashboards, KPI reporting, and manufacturing analytics to improve operational visibility and performance tracking.
- Drove NPI and manufacturing readiness activities by supporting technical RFQ (Request for Quote), capacity planning inputs, equipment specifications, FAT/SAT execution, and manufacturability requirements prior to production launch.
- Improved SoC EOL(end of line) test stand cycle time from 320 sec to 290 sec through process optimization, bottleneck reduction, and cross-functional execution.
- Participated in budget planning and cost analysis for equipment and tooling, MPRQ, CP/CPK (process capability and process capability index) and SPC, ensuring alignment with project goals and financial constraints.
- Identified operational and delivery risks early and executed mitigation strategies, reducing unplanned equipment downtime, performance data and rework by 10% year-over-year.
- Preserved product design integrity while resolving manufacturing constraints, enabling a 14% increase in production throughput without compromising quality.
- Drove supplier manufacturing readiness through capability assessments (MCA), process reviews, launch evaluations, and cross-functional coordination supporting successful NPI execution and production ramp-up.
- Conducted detailed cost analysis for retooling the entire station, identifying opportunities for cost reduction and process optimization, resulting in improved budget forecasting and operational efficiency.
- Worked managing the high production automated line, while running fast paced production-moving pallets, robots, and fixing the station without delaying the production loss.
- Established SPC monitoring and process performance tracking using manufacturing KPIs, capability analysis, and quality metrics to support production stability and yield improvements.
- Partnered with Original Design Manufacturers (ODM) during NPI programs to drive manufacturing readiness, capability assessments, process validation, FAT/SAT, capacity planning, and production launch execution.
- Managed third-party vendors and partners across multiple projects, meeting 100% contractual milestones and maintaining strong cross-team and vendor relationships valued around \$30million for quarterly.
- Led the design, build, installation, qualification, and launch of facilities and process-supporting equipment, enhancing operational capabilities for entire EDM, Stator & Rotor line.
- Designed custom leak test seals to improve part sealing and ensure consistent airflow during EDM operations, enhancing test accuracy and process reliability.
- Served as Subject Matter Expert (SME) in root cause analysis and corrective action planning, ordering the spare parts, Dc torque guns, resolving over 87% of quality issues within target timelines.
- Provide technical feedback concept architects, designers, test engineers and calibration teams to lead the process and improve the cycle & throughput time.
- Improved manufacturing system performance through process optimization, cycle time reduction, and production layout evaluation to enhance efficiency and capacity utilization.
- Conduct DFM/DFA meeting to troubleshooting the process, and build line retooling to improve the project efficiency.
- Collaborated with operators and team leads to ensure seamless product changes, factory change and successful execution of pilot builds, supporting launch readiness and minimizing disruptions to production flow.
- Supervised rework and repair operations, ensuring 100% compliance with engineering standards and maintaining product reliability across all units.
- Ensured all de-install and equipment handling activities followed safety, SMI105, EHS, and factory compliance standards, including proper documentation and sign-off before execution.
- Maintained a positive and respectful working relationship with union stewards, union employees, fostering a collaborative work culture that enhanced team morale and operational efficiency.
- Collaborated across departments from shop floor technicians to senior management to implement continuous improvement initiatives and standardize manufacturing practices across multiple facilities.
- Maintained effective communication with union representatives, Trades and contract manufacturers(HMH, J&J, COMAU, KUKA, FANUC) to foster strong working relationships and proactively resolve potential conflicts.

- Utilized fundamental engineering concepts and tools to analyze operational data and methods within field operations, achieving an efficiency rate of 90%.
- Formulated issues logically and derived comprehensive results to drive operational enhancements and cost control measures, leading to a cost reduction of 12%.

Refusal Survey Interviewer, IDPH NIU Survey Research Laboratory, DeKalb, IL October 2022 - December 2023

- Administered surveys and interviews with participants, ensuring accurate data collection for research programs.
- Coordinated with the team to ensure data accuracy, confidentiality, and adherence to ethical guidelines.

Operational Engineer, PAR Engineering Pvt Ltd, India June 2018 - December 2021

- Facilitated engineering projects supporting plant operations, process improvements, and new tooling installation, outlining and maintaining accountability for standard operating procedures (SOPs).
- Utilized PFMEA to analyze processes, identifying bottlenecks for cost and time reduction by 23% cycle time.
- Implemented Lean and Six Sigma methodologies, achieving a 15% increase in overall operational efficiency.
- Achieved significant efficiency gains and reduced labor costs through robotic automation by 30% efficiency.

Project Engineer, Shan Enterprises, India September 2016 - May 2018

- Collaborated with cross-functional teams, achieving a 20% decrease in post-sales issues by enhancing quality through enhancing quality standards.
- Implemented improvements resulting in enhanced product durability and reduced production defects by 41%.

Plant Supervisor, Shakthi Engineering Company, India April 2012 - March 2013

- Employed 3D CAD for plan creation, ensuring accurate manufacturing processes by guiding staff effectively, leading to resulting in streamlined production and reduced errors.
- Managed the PDCA and FEMA activity to improve the process and minimize the waste.

Welder/MC Operator, Shakthi Engineering Company, India January 2010 – April 2012

- Executed efficient GMAW/MIG welding for diverse projects, ensuring structural integrity.
- Operated CNC machines to perform precision execution and process discipline machining on micro-scale tasks, achieving tolerances.

PROFESSIONAL ORGANIZATIONS

Society of INFORMS, NIU Chapter September 2022 - December 2023

- ***Vice President***
 - Partnered with industry professionals to facilitate industrial visits for students, offering valuable insights.
 - Organized mentors for speeches and supported students in overcoming public speaking fears.

Society of Automotive Engineers, November 2013 - May 2016

- Led the design of a BAJA vehicle, showcasing innovative engineering solutions for competitions.
- Collaborated closely with production teams to implement effective foam repair processes, achieving a 100% success rate in material restoration.

Volunteering Experience

- Habitat for Humanity of DeKalb County: Construction projects for needy families.
- Feed My Starving Children, Illinois: Meal packing for hungry children.